

ColorCraft Paint

A modern, interactive paint-company website featuring a WebGL fluid background, virtual color try-on, colour harmony tools, and a paint quantity calculator.

PROJECT	TYPE	PREPARED	STATUS
paint-site	Single-Page Web App (SPA)	September 2026	Build & lint passing

1. Overview

ColorCraft Paint is a single-page marketing and product website for a paint company. It lets customers browse a catalogue of paint colours, preview them on a virtual room, generate complementary colour palettes, estimate how much paint a room needs, and book a free site visit.

All content is client-side and requires no backend or database — the entire experience runs in the browser.

2. Technology Stack

The site is built primarily with **JavaScript (ES2023+ with JSX syntax)** on top of the **React** UI library, bundled by **Vite**.

Area	Library / Tool	Version	Purpose
Language	JavaScript (ESM + JSX)	—	Application logic & components
UI Library	React	19.2.8	Component-based interface
DOM Renderer	React DOM	19.2.8	Renders React to the browser
Build Tool	Vite	^8.3.0	Dev server & production bundling
React Plugin	@vitejs/plugin-react	^6.1.1	Fast Refresh & JSX transform
Styling	Tailwind CSS	^4.3.3	Utility-first CSS framework
Tailwind Vite plugin	@tailwindcss/vite	^4.3.3	Tailwind integration with Vite
Animation	Framer Motion	^13.4.0	Declarative UI animations
3D / WebGL	Three.js	^0.186.0	GPU shader background
3D for React	@react-three/fiber	^9.7.0	React renderer for Three.js
Icons	lucide-react	^1.47.0	SVG icon set
Linting	oxlint	^1.81.0	Fast JavaScript linter

Shader language: The animated background is written in **GLSL** (OpenGL Shading Language) — custom vertex and fragment shaders executed on the GPU via WebGL through Three.js.

Languages used at a glance

- JavaScript / JSX — main application
- GLSL — WebGL shaders
- HTML — document shell
- CSS (Tailwind) — styling

3. How It Is Built

Development

```
npm install
npm run dev
```

Starts Vite dev server at `http://localhost:5173` with hot reload.

Production

```
npm run build
npm run preview
```

Outputs optimised static files to the `dist/` folder, ready to deploy to any static host.

Quality check: `npm run lint` runs oxlint — currently **0 warnings, 0 errors**. The production build completes successfully.

4. Project Structure

```

paint-site/
├─ index.html           # App shell, fonts, meta tags
├─ package.json         # Dependencies & scripts
├─ vite.config.js       # Vite + React + Tailwind config
├─ public/
│   └─ paint.svg        # Favicon
└─ src/
    ├─ main.jsx         # React entry point
    ├─ index.css        # Tailwind theme, keyframes, global styles
    ├─ App.jsx          # Layout & global state
    ├─ components/
    │   ├─ PaintBackground.jsx # WebGL fluid shader background
    │   ├─ CursorTrail.jsx    # Paint-splash cursor effect
    │   ├─ Navbar.jsx        # Sticky nav + scroll progress
    │   ├─ Hero.jsx          # Hero + rotating words
    │   ├─ ColorExplorer.jsx  # Searchable colour catalogue
    │   ├─ RoomPainter.jsx    # Virtual try-on, finishes, harmony
    │   ├─ PaintCalculator.jsx # Paint quantity & cost estimator
    │   ├─ Testimonials.jsx   # Scrolling review marquee
    │   ├─ Services.jsx      # Service cards
    │   ├─ FAQ.jsx           # Accordion
    │   ├─ Contact.jsx       # Booking form
    │   ├─ Footer.jsx        # Footer
    │   ├─ SectionHeading.jsx # Reusable heading
    │   └─ Toast.jsx         # Notification system
    └─ data/
        └─ colors.js        # 48 paint colours, categories, finishes
    └─ utils/
        └─ helpers.js       # Colour maths, clipboard, class helper

```

5. Key Features

- **Reactive WebGL background** — a flowing liquid-paint animation written as a custom GLSL fragment shader with domain-warping (fBm noise). It smoothly re-tints itself to match the colour the user selects.
- **Paint-splash cursor trail** — a canvas layer drops coloured blobs that fade as the pointer moves (disabled on touch devices and for reduced-motion users).
- **Colour catalogue** — 48 curated colours across 6 categories (Warm, Cool, Neutral, Bold, Pastel, Earth). Search by name, hex code, or colour family.
- **Virtual room try-on** — preview any colour on a stylised room with selectable finish (Matte / Satin / Gloss) and a Day / Evening lighting toggle.
- **Colour harmony** — automatically generates Complementary, Analogous, Triadic and Monochromatic palettes from the selected colour.
- **Tints & shades ramp** — shows lighter and darker variants of the chosen colour.
- **Paint calculator** — estimates paintable area, litres required, recommended pack sizes (20L / 4L / 1L) and approximate cost from room dimensions and number of coats.
- **Testimonials marquee** — two infinite, opposite-direction scrolling rows of customer reviews.
- **FAQ accordion** — animated expand/collapse answers.
- **Contact / booking form** — validated form with toast feedback.

- **Toast notifications** — reusable notification system for copy actions and form submissions.
- **Responsive & accessible** — mobile menu, keyboard-friendly controls, and reduced-motion support.

6. Component Reference

Component	Responsibility
<code>PaintBackground</code>	Full-screen fixed WebGL canvas running the animated fluid shader; accepts the selected colour as an accent.
<code>CursorTrail</code>	2D canvas overlay that paints fading colour blobs behind the pointer.
<code>Navbar</code>	Fixed navigation, mobile menu, and a scroll-progress bar.
<code>Hero</code>	Landing section with animated headline (rotating room names) and colour swatches.
<code>ColorExplorer</code>	Searchable, filterable grid of paint swatches with copy-to-clipboard.
<code>RoomPainter</code>	Virtual room preview, finish selector, day/evening mode, shade ramp and harmony palettes.
<code>PaintCalculator</code>	Estimates paint quantity, pack sizes and cost.
<code>Testimonials</code>	Animated review marquee.
<code>Services</code>	Cards describing interior/exterior painting, waterproofing and textures.
<code>FAQ</code>	Accordion of common customer questions.
<code>Contact</code>	Free site-visit booking form.
<code>Toast</code>	Context-based notification provider and hook.
<code>SectionHeading</code>	Consistent section titles and eyebrow labels.
<code>Footer</code>	Brand footer.

7. Data & Utilities

`src/data/colors.js`

Exports `COLORS` (48 entries, each with `name`, `hex`, `category` and `family`), `CATEGORIES`, `FINISHES` and a `DEFAULT_COLOR`.

`src/utils/helpers.js`

Colour-science and utility helpers, including:

- Conversions: `hexToRgb`, `rgbToHex`, `hexToHsl`, `hslToHex`
- Manipulation: `mix`, `rotateHue`, `adjustLightness`, `shadeRamp`, `harmony`
- Contrast: `getLuminance`, `isLight`
- Utility: `copyText`, `cn`

8. Design System

Token	Value	Usage
Background	#0c0f14	Base dark surface
Brand	#f59e0b → #ef4444	Primary gradient (amber → red)
Accents	#3b82f6 , #8b5cf6	Blue & violet highlights
Text	#f4f7fb / #9aa7b8	Primary & muted text
Display font	Space Grotesk	Headings
Body font	Inter	Body text

The dark theme with vivid gradients mirrors modern paint-brand websites and keeps foreground content readable over the animated background.

9. Performance & Accessibility

- **Code splitting** — Three.js is loaded lazily in a separate chunk, keeping the main bundle small (~386 KB; the WebGL chunk loads afterwards).
- **GPU animation** — the background runs entirely on the GPU via a shader, so it stays smooth without heavy CPU work.
- **Reduced motion** — animations slow or freeze for users who prefer reduced motion.
- **Touch-aware** — the cursor trail is disabled on touch devices.
- **Responsive** — layouts adapt from mobile to desktop; the navigation collapses into a menu.

10. References & Inspiration

Feature patterns were informed by leading paint-brand tools:

- **Sherwin-Williams ColorSnap Visualizer** — colour try-on and day/evening lighting.
- **Behr Color Visualizer** — “how much paint do I need?” and sheen selection.
- **PPG Paints Visualizer** — curated palettes and colour-family browsing.

11. Notes & Next Steps

- All colours, prices, phone numbers and testimonials are **sample/placeholder data** for demonstration and should be replaced with real business information.
- Possible extensions: room photo upload with automatic wall detection, physical colour-chip ordering, a backend for the booking form, and multi-language support.

